

Linear Algebra, Lecture 4

Eigenvalues and Eigenvectors

Autumn term

Outline

1. Definitions
2. The characteristic polynomial
3. Diagonalisation
4. References

Definition

Let A be an $n \times n$ matrix. A scalar L is an eigenvalue of A if there is a nonzero vector v with

$$A v = L v$$

Such a v is called an eigenvector of A for L .

The set of all such v , together with 0 , is the eigenspace.

The characteristic polynomial

$A v = L v$ holds for some nonzero v exactly when

$$\det(A - L I) = 0$$

The left hand side is a degree n polynomial in L ,
called the characteristic polynomial of A .

Its roots are precisely the eigenvalues of A .

Diagonalisation

If A has n linearly independent eigenvectors, collect them as the columns of P . Then

$$P^{-1} A P = D$$

where D is diagonal and holds the eigenvalues.

Powers become cheap: $A^k = P D^k P^{-1}$.

References

Strang, Introduction to Linear Algebra

Axler, Linear Algebra Done Right